ORF363/COS323
Linear Algebra and Calculus review (Lec2.pdf)

Ioannis Akrotirianakis

Today's topics

- Linear Algebra and Calculus refresher
- Inner products and norms
- Eigenvalues and eigenvectors
- Elements of differential calculus
- Taylor expansion

Waitlist is full

LINEAR ALGEBRA review

Inner products

Definition: The inner product of two vectors $x, y \in R^n$ is the scalar:

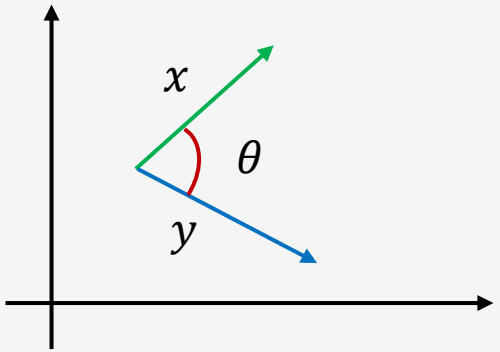
$$< x, y > = x^T y = \sum_{i=1}^{n} x_i y_i$$

Vectors

$$x = \begin{pmatrix} x_1 \\ \dots \\ x_n \end{pmatrix} \qquad x^T = (x_1 \quad .. \quad x_n)$$

Properties:

Positivity	$x^T x \geq 0, \ \forall x \in R^n$ and $x^T x = 0$ iff $x = 0$
Symmetry	$x^T y = y^T x$
Additivity	$(x + y)^T z = x^T z + y^T z$
Homogeneity	$(rx)^T y = r x^T y, \quad \forall r \in R$



Geometric interpretation: if $x, y \in R^2$ then $x^T y = length(x) \times length(y) \times \cos(\theta)$

Remark: the inner product is zero when $\theta = \frac{\pi}{2}$, positive when $\theta \in \left[0, \frac{\pi}{2}\right)$ and negative when $\theta \in \left(\frac{\pi}{2}, \pi\right]$

Orthogonality: two vectors $x, y \in R^n$ are called orthogonal iff: $x^T y = 0$

Example: $x = (1,2)^T$ and $y = (3, -\frac{3}{2})$ are orthogonal because $x^T y = 0$ (Question: what is their angle θ ?)

Norms → ||·||

Definition: It is a real valued functions $||\cdot|| : R^n \to R$ that satisfies the properties:

<u>Positivity:</u> $\left||x|\right| \geq 0$ and $\left||x|\right| = 0$ iff $x = 0$

<u>Homogeneity:</u> $\left||rx|\right| = |r|\,||x||\,,\ r \in R$

<u>Triangle inequality:</u> $\left||x+y|\right| \leq \left||x|\right| + \left||y|\right|$

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<u>Triangle inequality:</u> $||x + y|| \leq ||x|| + ||y||$

Basic norms:

1-norm: $||\cdot||_1 = |x_1| + |x_2| + \ldots + |x_n|$

2-norm: $||\cdot||_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$

∞-norm: $||\cdot||_\infty = \max\{|x_i| : i = 1,2,\ldots,n\}$

The inequality connecting the three norms: $||x||_\infty \leq ||x||_2 \leq ||x||_1\,,\ \forall x \in R^n$

Connection between <u>inner product</u> and 2-norm $||\cdot||_2 = \sqrt{x^T x}$

Cauchy-Schwartz inequality: $|x^T y| \leq ||x||\,||y||\,,\ \forall x, y \in R^n$

Norms → $||\cdot||$

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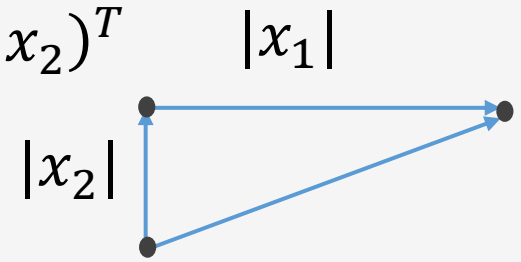
∞-norm: $||\cdot||_\infty = \max\{|x_i| : i = 1,2,\ldots,n\}$

$x = (x_1, x_2)^T$



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Connection between <u>inner product</u> and 2-norm $||\cdot||_2 = \sqrt{x^T x}$

Cauchy-Schwartz inequality: $|x^T y| \leq ||x||\,||y||\,,\ \forall x, y \in R^n$

Eigenvalues and Eigenvectors

Definition: Let $A \in R^{n \times n}$ be a square matrix. If $\exists \lambda \in R$ and $v \in R^n$ satisfying $\underline{Av = \lambda v}$
then λ is the eigenvalue and v is the eigenvector of A.

Computations:

for eigenvalues, solve the <u>characteristic polynomial</u>: $\det(\lambda I - A) = 0$
for the eigenvector corresponding to λ_i, solve the <u>linear system</u>: $Av = \lambda_i v$

Remark: If A is a <u>symmetric matrix</u>, then all its eigenvalues are <u>real numbers</u>

A square matrix $M \in R^{n \times n}$ is called **symmetric** iff $M^T = M$

Eigenvalues and Eigenvectors

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A square matrix $M \in R^{n\times n}$ is called **symmetric** iff $M^T = M$

Remark: all eigenvectors of A are orthogonal to each other → $v_i \perp v_j \Leftrightarrow v_i^T v_j = 0, \forall i, j = 1,2, \dots, n$ and $i \neq j$

Eigen-decomposition: $A = V\Lambda V^T = \sum_{i=1}^{n} \lambda_i v_i v_i^T$, where $V = [v_1, \dots, v_n] \in R^{n\times n}$ and $\Lambda = diag(\lambda_1, \dots, \lambda_n)$

Important note: You should be comfortable to compute eigenvalues and eigenvectors of **2×2 matrices**
For larger matrices we will ask you to use Matlab or Python.

Eigenvalues and Eigenvectors

Some other very useful properties:

(1) $\det(A) = \prod_{i=1}^{n} \lambda_i$

(2) We define the <u>Trace</u> of a matrix is the sum of its diagonal elements: $Tr(A) = \sum_{i=1}^{n} a_{ii}$

Connection of the Trace and eigenvalues: $\mathrm{Tr(A)} = \sum_{i=1}^{n} \lambda_i$

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

eigenvalues: $\lambda_1 = 16.1168440, \quad \lambda_2 = -1.11684397, \quad \lambda_3 = 0$

You can easily verify properties (1) and (2)

We will use the **Trace** of a matrix in the 2nd part of the course

Semidefinte Programming and **Sum of Squares Programming**

Positive (Semi) Definite matrices (P(S)D)

Definitions: A symmetric matrix $Q \in R^{n \times n}$ is said to be

<u>Positive SemiDefinite (PSD)</u>	$x^T Q x \geq 0$, $\forall x \in R^n$	→	$Q \succcurlyeq 0$
<u>Positive Definite (PD):</u>	$x^T Q x > 0$, $\forall x \in R^n$ & $x \neq 0$	→	$Q \succ 0$
<u>Negative SemiDefinite (NSD):</u>	$-Q$ is PSD	→	$Q \preccurlyeq 0$
<u>Negative Definite (ND)</u> iff	$-Q$ is PD.	→	$Q \prec 0$
<u>Indefinite</u>	it is neither PSD not NSD	→	$Q \not\succcurlyeq 0$

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Positive (Semi) Definite matrices (P(S)D)

Relationship between Definiteness of a matrix and the sign of its eigenvalues:

Theorem: A symmetric matrix Q is PSD $\Leftrightarrow$ all its eigenvalues are ≥ 0

Theorem: A symmetric matrix Q is PD $\Leftrightarrow$ all its eigenvalues are > 0

Theorem: A symmetric matrix Q is NSD $\Leftrightarrow$ all its eigenvalues are ≤ 0

Theorem: A symmetric matrix Q is ND $\Leftrightarrow$ all its eigenvalues are < 0

Remark: A matrix is indefinite iff it has positive and negative eigenvalues.

Examples

#1. Using the previous definitions, determine the type of matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Using the definitions of PSD and PD, we select any $x \in R^2$ and compute:

$$x^T A x = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 + 2x_1x_2 + x_2^2 = (x_1 + x_2)^2$$

Since $(x_1 + x_2)^2 \geq 0, \ \forall x_1, x_2,$ we have that A is Positive SemiDefinite (PSD or $A \succcurlyeq 0$)

Question: is A Positive Definite ($A \succ 0$) ??

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Question: is A Positive Definite ($A \succ 0$) ??

NO -- you need to prove this, by finding a nonzero vector $x \neq 0$ such that $x^T A x = 0$

Consider for example the vectors: $x = (1, -1)^T$ or $y = -x$

Examples

#2. Using the previous definitions, determine the type of matrix $A = \begin{bmatrix} 5 & 2 \\ 2 & -3 \end{bmatrix}$

Observation: it contains positive and negative diagonal elements…

<u>Is it PSD ?</u>

<u>Is it NSD ?</u>

Examples

#2. Using the previous definitions, determine the type of matrix $A = \begin{bmatrix} 5 & 2 \\ 2 & -3 \end{bmatrix}$

Observation: it contains positive and negative diagonal elements…

<u>Is it PSD ?</u>

NO - select for example $x = (0,1)^T$ and compute: $x^T A x$ (it is $-3 < 0$)

<u>Is it NSD ?</u>

NO – select for example $y = (1,0)^T$ and compute: $y^T A\, y$ (it is $5 > 0$)

<u>Conclusion</u>: the matrix is indefinite ($A \ngeq 0$)

Positive (Semi) Definite matrices (P(S)D)

$$Q = \begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{14} & q_{15} \\ q_{21} & q_{22} & q_{23} & q_{24} & q_{25} \\ q_{31} & q_{32} & q_{33} & q_{34} & q_{35} \\ q_{41} & q_{42} & q_{43} & q_{44} & q_{45} \\ q_{51} & q_{52} & q_{53} & q_{54} & q_{55} \end{bmatrix}$$

<u>Sylvester's criteria</u>

#1: A symmetric matrix $Q \in R^{n \times n}$ is **PD** $\Leftrightarrow$ $\det(\Delta_1), \ldots, \det(\Delta_n) > 0$

Δ_i is the i-th **leading principal minor**

$$Q = \begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{14} & q_{15} \\ q_{21} & q_{22} & q_{23} & q_{24} & q_{25} \\ q_{31} & q_{32} & q_{33} & q_{34} & q_{35} \\ q_{41} & q_{42} & q_{43} & q_{44} & q_{45} \\ q_{51} & q_{52} & q_{53} & q_{54} & q_{55} \end{bmatrix}$$

Δ_1 Δ_2 Δ_3 Δ_3 $\Delta_5 = Q$

#2: A symmetric matrix $Q \in R^{n \times n}$ is **PSD** $\Leftrightarrow$ $\det(\Gamma_1), \ldots \det(\Gamma_{2^n - 1}) \geq 0$

Γ_i is the i-th **principal minor**

A principal minor of Q is computed by choosing a subset of rows and the same subset of columns of Q

$$Q = \begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{14} & q_{15} \\ q_{21} & q_{22} & q_{23} & q_{24} & q_{25} \\ q_{31} & q_{32} & q_{33} & q_{34} & q_{35} \\ q_{41} & q_{42} & q_{43} & q_{44} & q_{45} \\ q_{51} & q_{52} & q_{53} & q_{54} & q_{55} \end{bmatrix}$$

$$\Gamma_{1,3,4} = \begin{bmatrix} q_{11} & q_{13} & q_{14} \\ q_{31} & q_{33} & q_{34} \\ q_{41} & q_{43} & q_{44} \end{bmatrix}$$

Examples

Consider the 3×3 symmetric matrix $A = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$

Use Sylvester's criteria to determine the **conditions** (inequalities) that ensure it is PSD and PD.

Conditions for PD:

$$a > 0$$
$$ad - b^2 > 0$$
$$\det(A) > 0$$

Conditions for PSD:

$$a \geq 0, \quad d \geq 0, \quad f \geq 0,$$
$$ad - b^2 \geq 0, \quad af - c^2 \geq 0, \quad df - e^2 \geq 0$$
$$\det(A) \geq 0$$

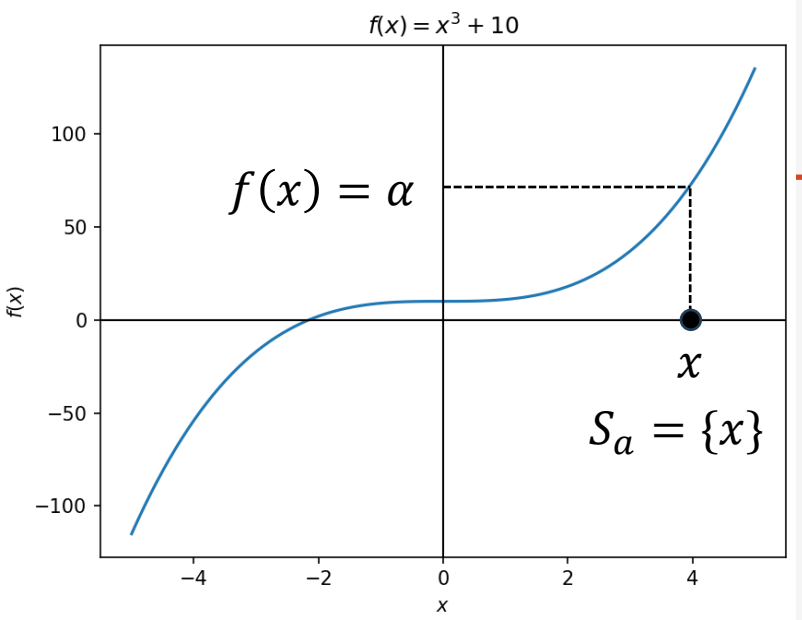
CALCULUS review

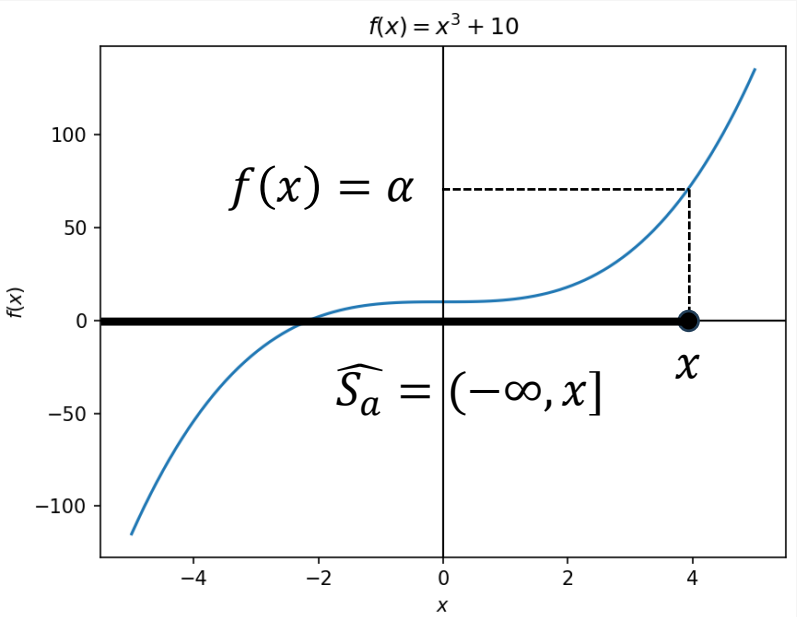
Commonly used classes of functions

- $f: R^n \to R$
- **Linear**: $f(x) = c^T x = \sum c_i x_i$, where $c \in R^n$
- **Affine**: $f(x) = c^T x + b = \sum c_i x_i + b$, where $c \in R^n$, $b \in R$
- **Quadratic**: $f(x) = x^T Q\, x + \ c^T x + b = \sum_i \sum_j x_i q_{ij} x_j + \sum_i c_i x_i + b$, where Q $\in R^{n \times n}$, $c \in R^n, b \in R$ and Q symmetric
- Exercise: write $f(x) = 3x_1^2 - 5x_1 x_2 + 4x_2 x_3 + 7x_1 - 9$ as $x^T Q\, x + \ c^T x + b$
  - You are looking for the (symmetric) matrix Q, the vector c, and scalar b

α-Level set & α-Sublevel set

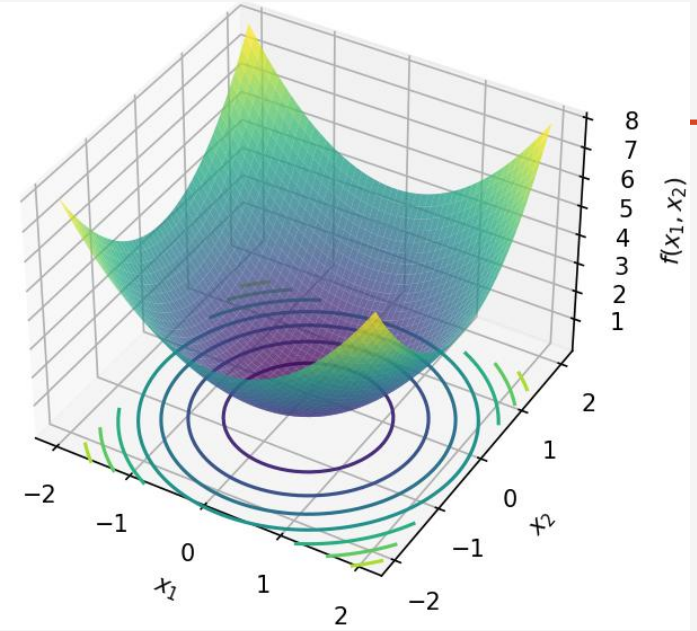
- $f: R^n \to R$
- **α-Level set** of the function f:
  - $S_a = \{x \in R^n | f(x) = \alpha\}$
- **α-Sublevel set** of the function f:
  - $\widehat{S_a} = \{x \in R^n | f(x) \leq \alpha\}$





α-Level & α-Sublevel sets

$\boldsymbol{f(x_1, x_2) = x_1^2 + x_2^2}$



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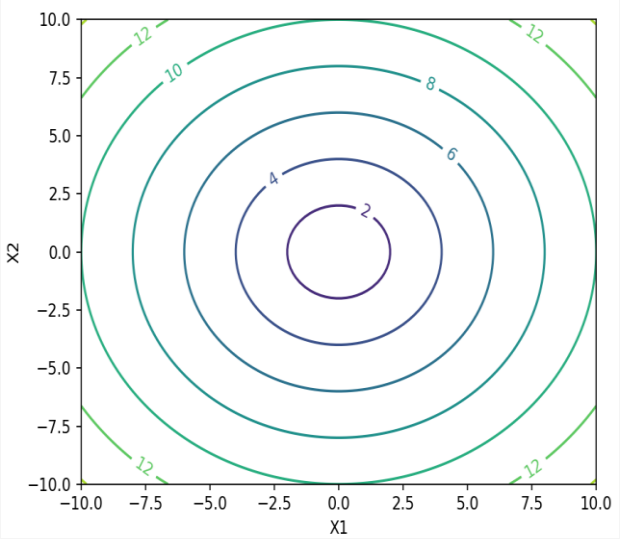
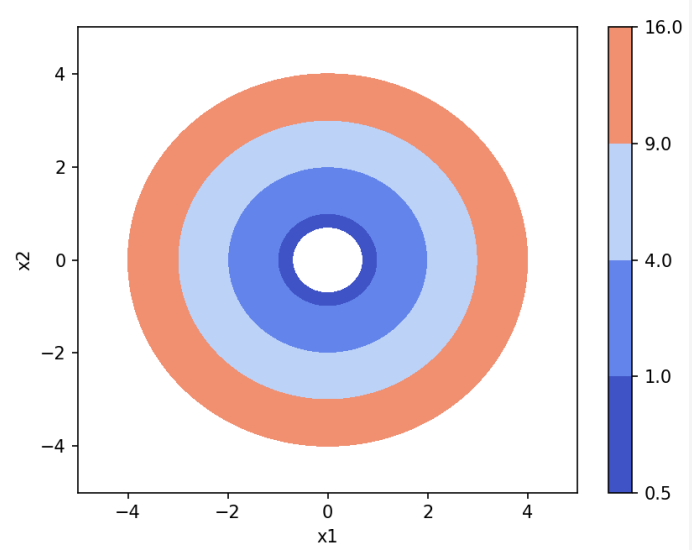
Level sets



Sublevel sets



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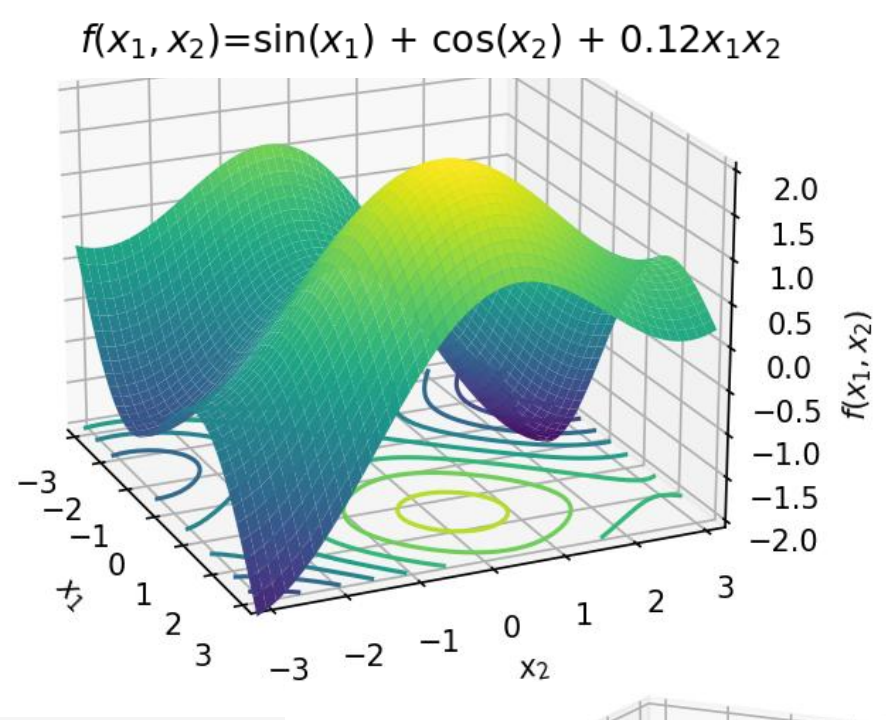
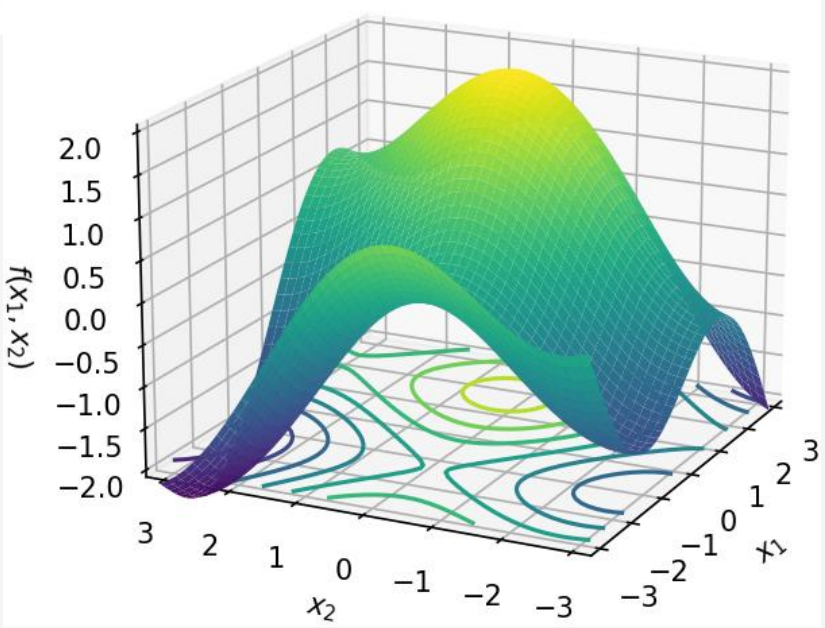
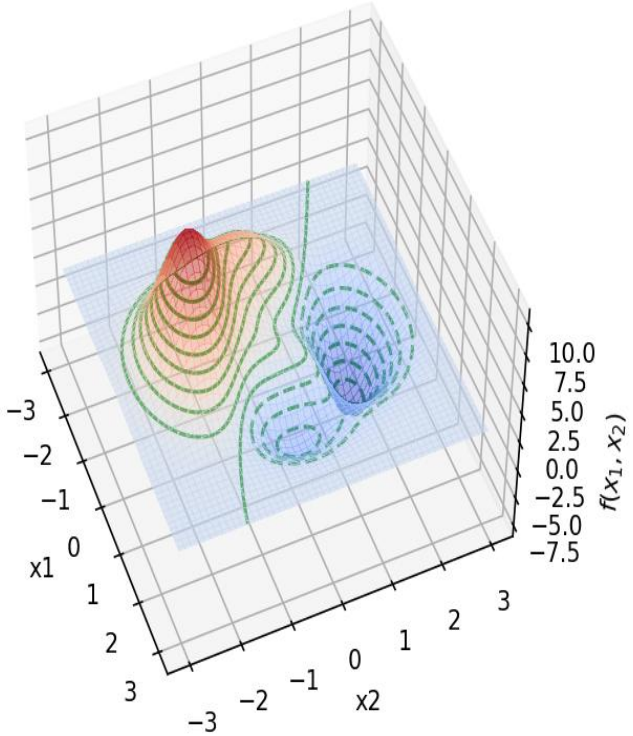
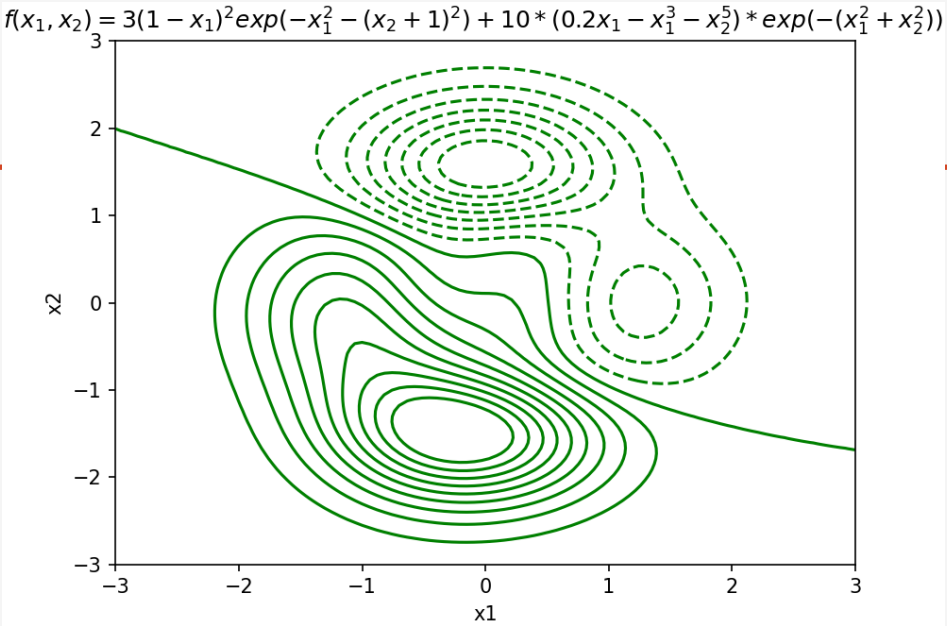




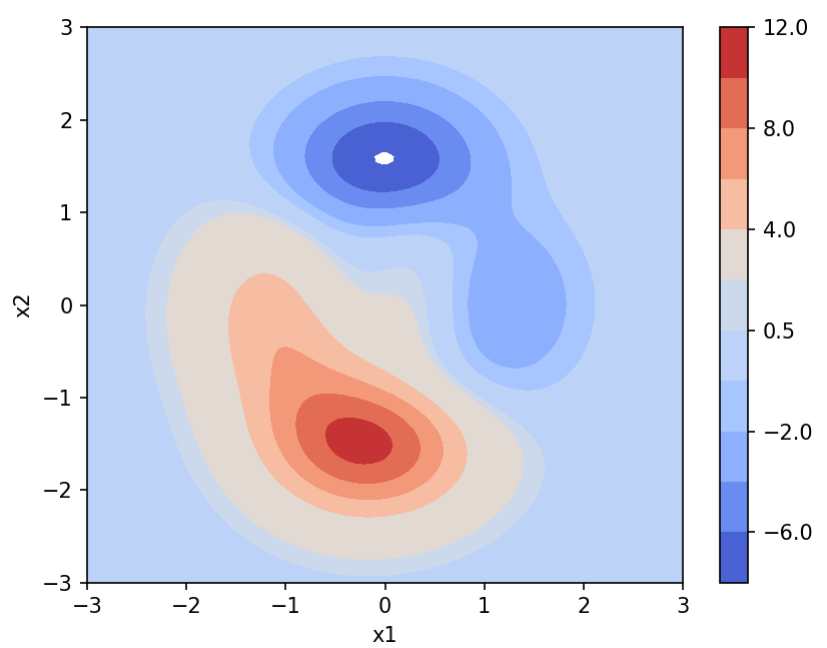
α-Level sets and α-Sublevel sets

$f(x_1, x_2) = 3(1-x_1)^2 exp(-x_1^2 - (x_2+1)^2) + 10*(0.2x_1 - x_1^3 - x_2^5)*exp(-(x_1^2 + x_2^2))$





Level sets



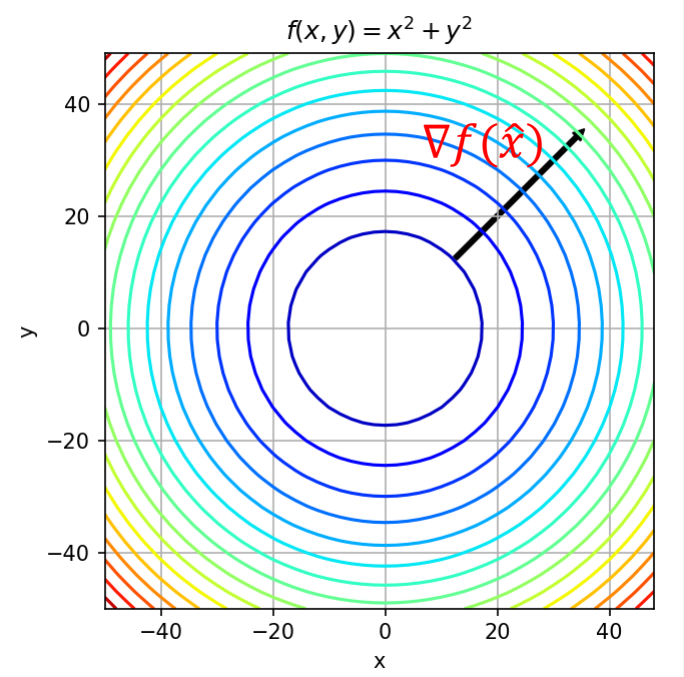
Sublevel sets

Gradient of a function

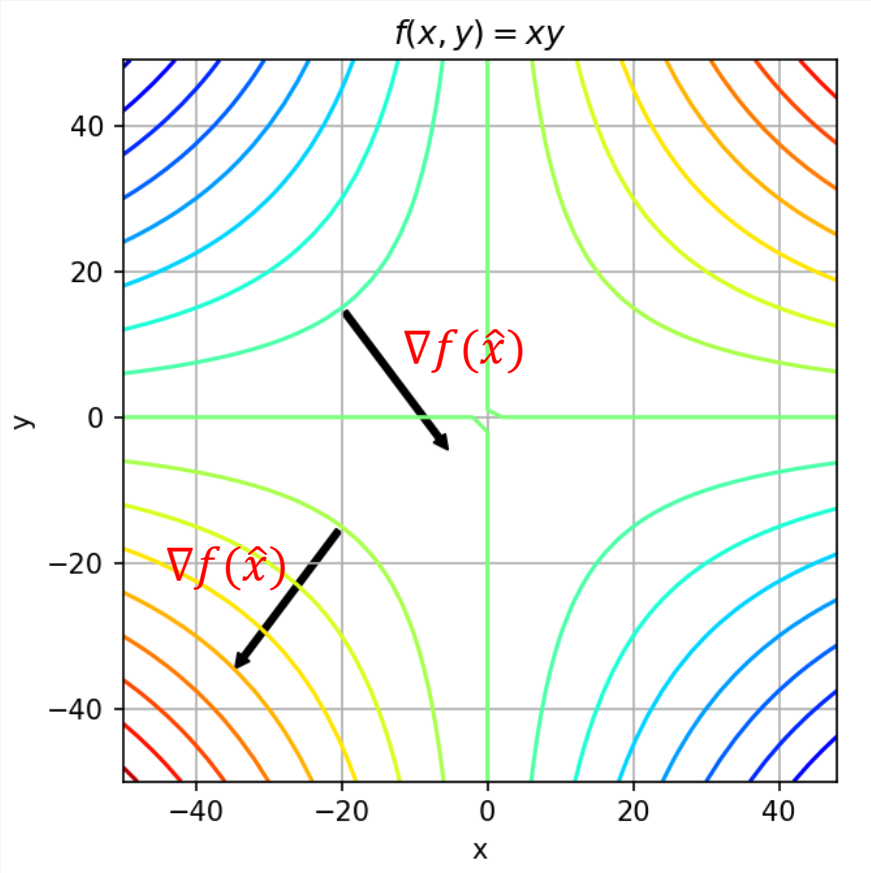
- $f: R^n \to R$, continuously differentiable
- $\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n}\right)^T \in R^n$
- $\frac{\partial f(x)}{\partial x_i}$ measures the **rate of change** of f along the coordinate i
- The gradient of f collects all those rates of change in one vector

Gradient of a function

- The gradient of f at a point $\hat{x}$ (i.e., the vector $\nabla f(\hat{x})$) tells us the **direction** where the function **increases most**
  - Obviously, $-\nabla f(\hat{x})$ tells us the direction where the function **decreases most**
  - **This is very useful in optimization**
- $\nabla f(\hat{x})$ is **perpendicular** to the tangent of the level set that passes through $\hat{x}$



Examples


$f(x, y) = xy$
$\nabla f(\hat{x})$
$\nabla f(\hat{x})$
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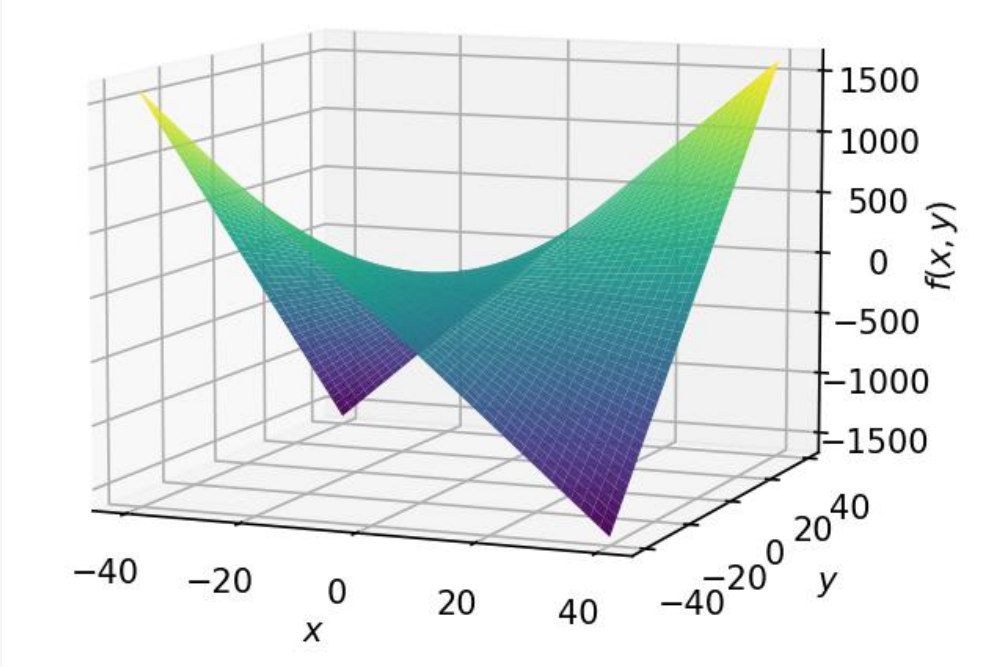
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$f(x, y)$
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x
−40
−20
0
20
40
y

Hessian matrix

Consider the function $f: R^n \to R$ to be twice continuously differentiable

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix} \in R^{n \times n}$$

Symmetric matrix because:

$$\frac{\partial^2 f(x)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x)}{\partial x_j \partial x_i}$$

The Hessian matrix contains information about the **curvature** of the function

Or in other words, the **rate of change of the partial derivatives** of a function

Examples of gradients and Hessians

- What are the gradient and Hessian of an affine function??
  - $f(x) = c^T x + b$
  - $\nabla f(x) = ??$ $\quad \nabla^2 f(x) = ??$
- What are the gradient and Hessian of a quadratic function??
  - $f(x) = x^T Q x + c^T x + b$
  - $\nabla f(x) = ??$ $\quad \nabla^2 f(x) = ??$

Chain rule

- You have two function: $f: (a, b) \to \underline{D \subseteq R^n}$ and $g: D \to R$

  - Note that $f(t) = \begin{pmatrix} f_1(t) \\ \dots \\ f_n\ (t) \end{pmatrix}$

- The **composite function**: $h: (a, b) \to R$ is defined as: $h(t) = (g \circ f)(t) = g(f(t))$

- The **gradient** of the composite function: $h'(t) = \nabla g\big(f(t)\big)^T \begin{pmatrix} f'_1(t) \\ \dots \\ f_n'(t) \end{pmatrix}$

- **Special case:** When $f(t) = tc + d$, for some $c, d \in D \subseteq R^n$, then: $h'(t) = \nabla g\big(f(t)\big)^T$ c

Chain rule – Example!!

- $f: (a, b) \to D \subseteq R^2$ with $f(t) = (f_1(t), f_2(t))^T = (e^t, t^2 + 1)^T$
- $g: D \to R$ with $g(x_1, x_2) = x_1^2 + x_2^3$
- $h: (a, b) \to R$ with $h(t) = (g \circ f)(t) = g(f(t))$
- $h'(t) = \nabla g\big(f(t)\big)^T \begin{pmatrix} f_1'(t) \\ f_2'(t) \end{pmatrix} = (2e^t, 3(t^2 + 1)^2) \begin{pmatrix} e^t \\ 2t \end{pmatrix} = 2e^{2t} + 6t(t^2 + 1)^2$

Remark:
You can also compute: $h(t) = g\big(f(t)\big) = \cdots = e^{2t} + (t^2 + 1)^3$ and then take the derivative wrt t

Taylor expansions

- What is all about?
  - A **local approximation** of a function around a specific point using a polynomial
  - We can **increase the degree** of the polynomial and obtain **better local approximations**
- Polynomials of degree 1 (**affine functions**) and degree 2 (**quadratic functions**) are frequently used in Optimization to **locally approximate nonlinear functions** (objectives or constraints)

Taylor expansions in R

- **Taylor's theorem**: Given a function $f: R \to R$ which is <u>m times continuously differentiable</u>, and an interval $[a, b] \subseteq R$. Then

Polynomial of degree m

- $\hat{f}(b) = \boxed{f(a) + f'(a)(b-a) + \frac{1}{2}f''(a)(b-a)^2 + \cdots + \frac{1}{m!}f^{(m)}(b-a)^m \quad +}$

$\boxed{+o(|b-a|^m)}$

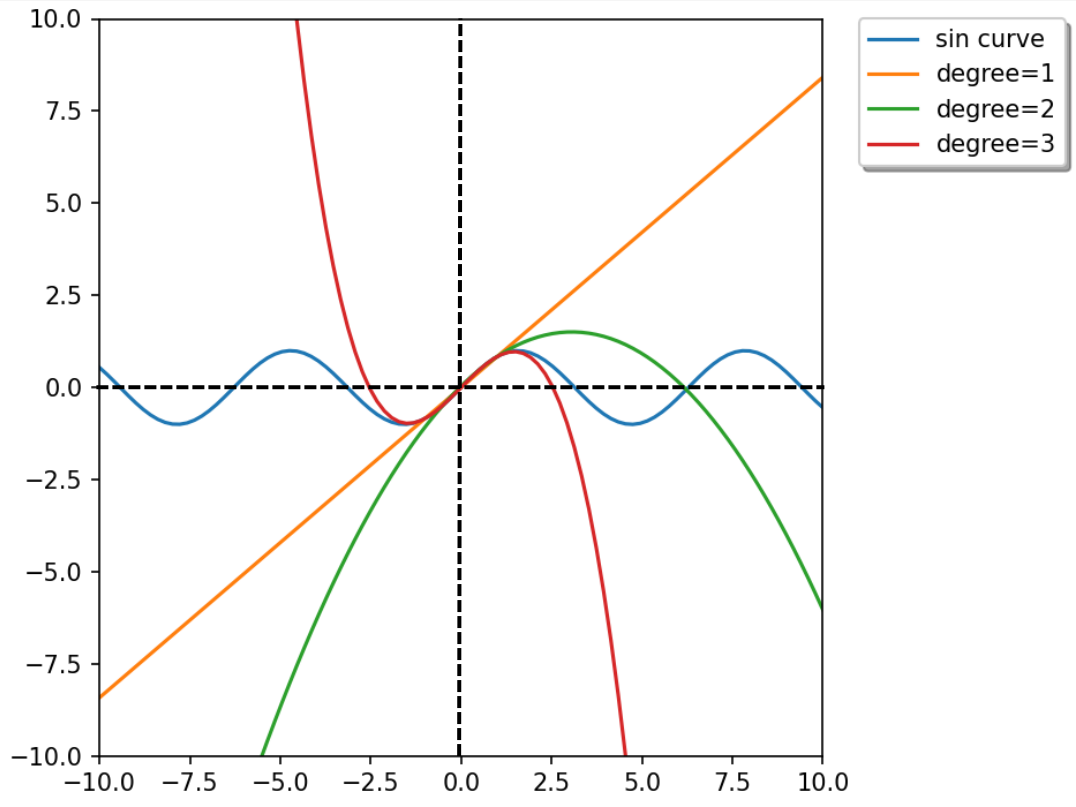
- **Error**: $\mathrm{err(b)} = f(b) - \hat{f}(b) = f(b) - \left(f(a) + f'(a)(b-a) + \frac{1}{2}f''(a)(b-a)^2 + \cdots + \frac{1}{m!}f^{(m)}(b-a)^m \right)$

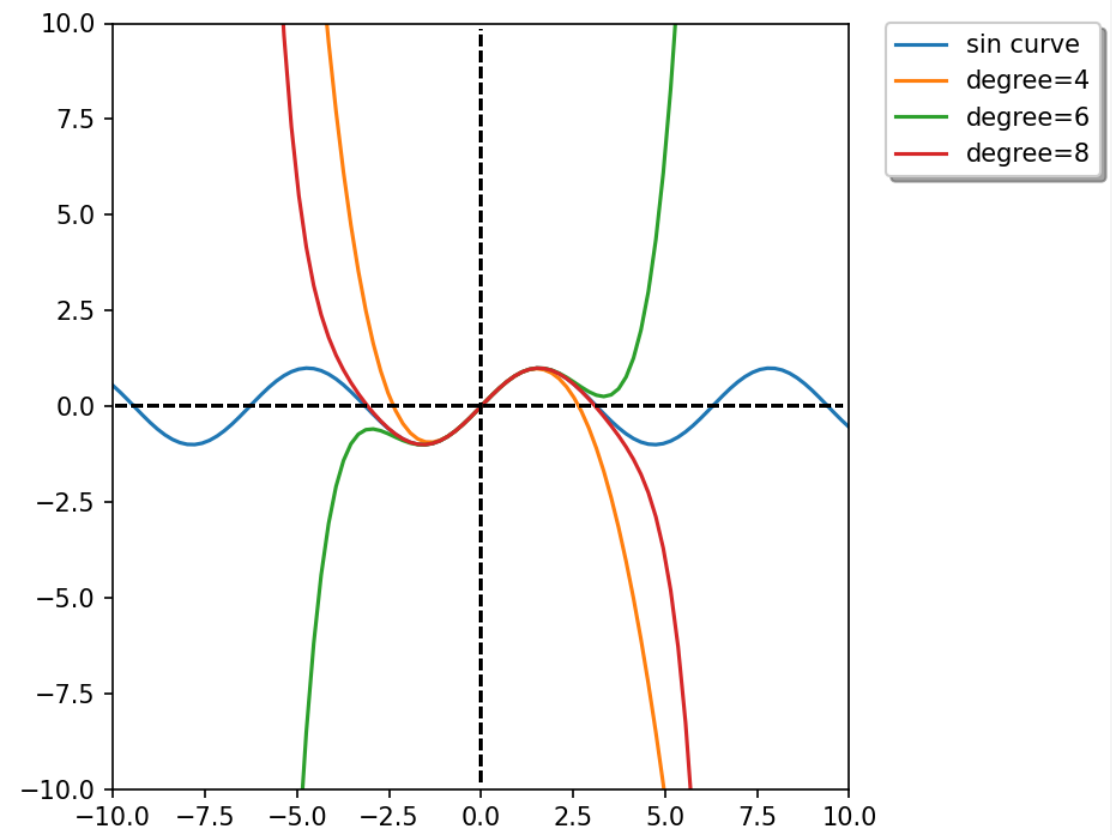
- $o(|b-a|^m) = \lim_{b \to a} \left(\frac{|err(b)|}{|b-a|^m} \right) \quad = 0$

- **In English:** "the error of the m-th order Taylor approximation goes to 0 faster than $|\mathrm{b} - a|^m$

Examples for Taylor approximations

F(x)=sin(x)





The original function $f(x) = \sin(x)$ and some of its Taylor approximations at $x_0 = 0$

Taylor approximation in R^n

- Given $f: R^n \to R$ twice continuously differentiable
- *<u>1st order Taylor approx</u>. (what type of function is it?)*
  - $f(x) = f(x_0) + \nabla f(x_0)^T (x - x_0) + o\left(\left|\left|x - x_0\right|\right|\right)$
- *<u>2nd order Taylor approx</u>. (what type of function is it?)*
  - $f(x) = f(x_0) + \nabla f(x_0)^T (x - x_0) + \frac{1}{2!}(x - x_0)^{\mathrm{T}} \nabla^2 f(x_0)(x - x_0) + o\left(\left|\left|x - x_0\right|\right|^2\right)$
- **Taylor is in your HW1**

Taylor approximation in R^n

- Given $f: R^n \to R$ twice continuously differentiable
- *1st order Taylor approx. (Convert it to* $\underline{c^T x + b}$*)*
  - $f(x) = f(x_0) + \nabla f(x_0)^T (x - x_0) + o\left(||x - x_0||\right)$
- *2nd order Taylor approx. (Convert it to* $\underline{x^T Q x + c^T x + b}$ *)*
  - $f(x) = f(x_0) + \nabla f(x_0)^T (x - x_0) + \frac{1}{2!}(x - x_0)^{\mathrm{T}} \nabla^2 f(x_0)(x - x_0) + o\left(||x - x_0||^2\right)$
- **Taylor is in your HW1**

Exercise

- Find the first 3 orders of Taylor approximations for $f(x) = e^x$ at $\hat{x} = 0$
- $f_0(\hat{x}) = ??, f_1(\hat{x}) = ??, f_2(\hat{x}) = ??, f_3(\hat{x}) = ??$
- What type of polynomials are these??
- What is the error in each of the above cases??

Appendix

Square matrices – Symmetric matrices – PSD matrices – *w.l.o.g.*

- When we have a square matrix $A \in R^{n \times n}$ that is PSD, PD, NSD, ND, the matrix can be considered to be symmetric, **without loss of generality (*w.l.o.g.*).**

- <u>Why??</u>

- I can always write $A = \frac{1}{2}\boxed{(A + A^T)} + \frac{1}{2}\boxed{(A - A^T)}$

  - $A + A^T$ is symmetric matrix. <u>Why?</u>

    - Define $\mathrm{B} = A + A^T$. Then $B^T = (A + A^T)^T = A^T + A = A + A^T = B$.

  - $x^T(A - A^T)x$ is always 0, for every $x \in R^n$. <u>Why?</u>

    - Note this: $x^T A x = x^T z = z^T x = (Ax)^T x = x^T A^T x \Rightarrow x^T A x$- $x^T A^T x = 0 \Rightarrow x^T(A - A^T)x = 0$

      (I just define $z = Ax$)